
SITE VISIT REPORT

Client: Länsförsäkringar	Project: Architecture Review	Oracle Project Code:
Oracle Consultant: Jomon Kunnel	Date(s) of work covered by this report: 08-Oct-2018 to 19-Oct-2018	Total Billable Time: 10 Days Billable

Architecture and Configuration Review

Over the past 10 days we went over the current production and new production environment of Oracle Goldengate configuration. We focused the production system. These are the conclusions and recommendations:

Platform Certification

Currently the system is running on Wintel 64-bit platform with Oracle database 12c version 12.1.0.2.0 with compatibility is set to 12.1.0.2.0. SGA values are set to 35G. The database configured on 2 node RAC

Security

Wallets

Wallets are used in current environment to maintain the security on Goldengate side. Credential store is used as a best practice.

Macros

Macros are used to keep the table mapping away from the par files, and this also helps to keep the configuration tidy.

Components

Servers

Wintel servers are used in the current production environment. It is observed the performance of database servers is not optimal. It is also observed that the server does take more than 25 minutes to reboot, if required, and I/O is also not doing a good performance. Interconnect network between RAC nodes are not doing good, this need to be re-assessed, and recalibrated.

Recommendation:

1. Storage layer needs tuning.
2. Further expert level investigation is recommended on network configuration.
3. Hardware level calibration is required database servers

Folders

ACFS is used for keeping Goldengate binaries, and trail files. This makes the files available across the RAC nodes.

Configuration

Goldengate Manager

Manager configuration has following entry in it

```
PORT 7827
DynamicPortList 20001-20099
PurgeOldExtracts H:\Goldengate\dir\*, UseCheckPoints,
MinKeepHours 72
AUTOSTART ER *
```

This means the configuration has a non-default port to communicate and a dynamic port list is provided. The configuration does the housekeeping by deleting old trail files based on the checkpoint details. The files will be kept for 72 hours.

Globals

Global parameter file is kept on clusterware and it has following entries in it

```
GGSHEMA ggadmin
ALLOWOUTPUTDIR H:\Oracle\Goldengate\dir\
checkpointtable ggadmin.gg_checkpoint
```

TRANDATA

Trandata is enabled on every table, however this should be done on all columns of the tables, because the business needs every updates converted to an insert on target table.

Recommendation: do dblogin on source, and execute following for individual tables

```
dblogin
add trandata schema.tablename allcols
```

OR

```
add trandata schema.* allcols
```

Also make sure the trandata is added to every new table before adding them to the mapping.

Data Verification

It is observed that there is a manual verification of transactions in place; this might be inadequate to find any data loss due to any exceptions configured on gg to handle data related errors.

Oracle Goldengate Veridata can be a tool to be considered for data comparison, and it has the capability to identify data inconsistency, and it has the capability of working on live databases with constantly changing data

Database

An observation into the database tells that the current production servers are not doing an optimum performance. There is a combination of problems on the database, which included network/storage and OS level problems. It indicates that the database may not be able to handle the production load.

Checkpoint

Goldengate is using checkpoint table ggadmin.gg_checkpoint. Replicat uses the table to maintain a record of its read position in the trail for recovery purposes.

Log Entries / Problems found

Missing transactions

Business requirement is to convert every update statements into an insert on target, and this will create a new row on target table.

It is noticed that some of the update statements happening on source database go missing in target database, and this is because values from all columns of the row is not made available on target. The solution is to add trandata to all columns.

Recommendation: add trandata schema.tablename allcols

OR

add trandata schema.* allcols

Cannot find the parameter 'INSERTUPDATES'.

Business requirement is to convert all update statements on source database to be converted into an insert on target.

It is recommended to have INSERTUPDATES on the replicat parameter file before the table mapping.

DDL replication is not enabled.

DDL replication is not enabled, and recommended to execute following scripts to enable it

@role_setup.sql

grant ggs_ggsuser_role to ggadmin;

@ddl_enable;

@ddl_pin ggadmin;

Recyclebin is not disabled.

Best practice recommendation is to disable recyclebin of the database, especially when there are DDL replications in place.

Force_logging is not disabled.

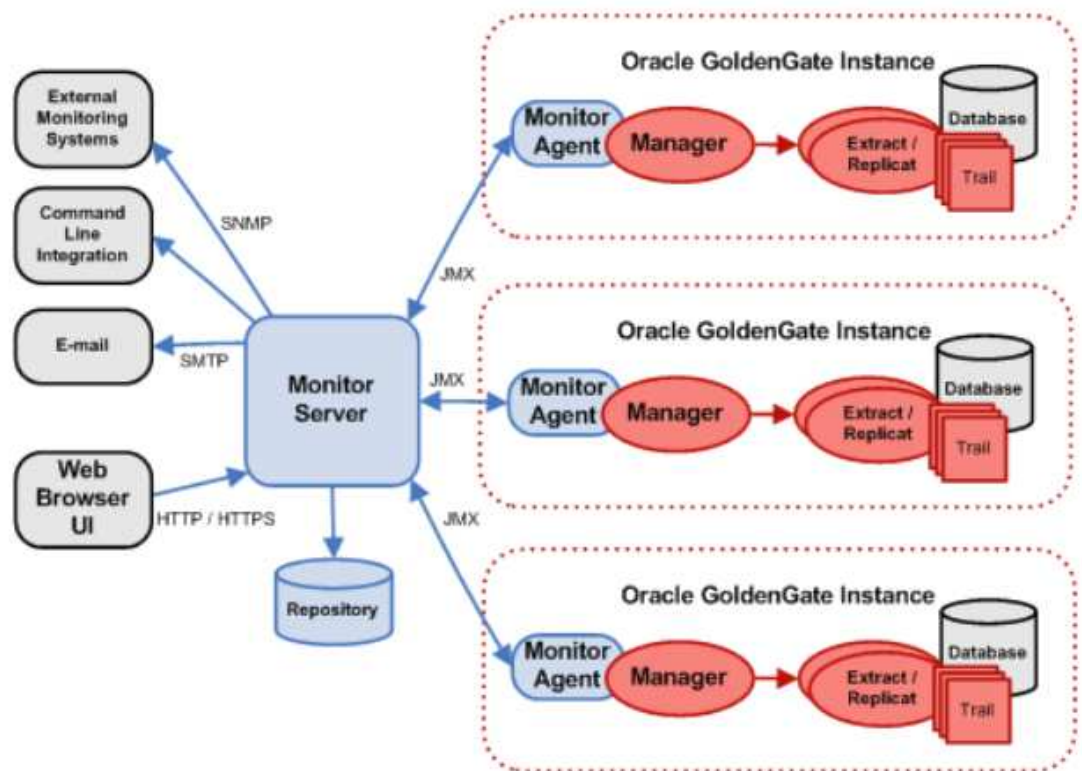
This will make the transaction not being captured if a job executes with no logging.

Recommendation: Enable force_logging on database

Monitoring

There are no automated monitoring processes in place.

Recommendation : Use a monitoring tool such as Goldengate Monitor. An illustration of architecture is as following:



Other checks

IP Usage policy

A discussion with the solution architect made me aware that the use of IPV6 is against the policy, however Goldengate is using IPV6 because it is made available.

Recommendation: Add following to globals parameter file and restart the processes

USEIPV4

DDL on objects

It is observed that the structural changes are introduced on production database objects without DBA awareness. This will make unexpected problems.

Recommendation: Is to make every change through a proper change management process and with consultation of DBA and approval.

Admin Accounts

It is observed that admin users on server are not created with a production standard. A standard template for account creation is recommended for creating OS users for DBA accounts.

DW and OLTP on same database

Database is configured as OLAP. This will impact the performance of production system. As a best practice, the production database should be OLTP and DW should be on OLAP.

Conclusion

It is observed that, the Goldengate configuration at Lansforsakringar was done successfully and made the replication running but limited to DML. It appear that there were some crucial steps were missing, or ignored, it is highly possible that the then business requirement did not have a requirement of DDL replication. Recommendations for the missing steps and configuration are made on this document along with the findings.

New production environment on Linux OS is appeared to be stable and better configured from the database/GG point of view. However it is recommended to do a hardware/OS/Network/storage assessment is done on it before going live on it.

OLAP database is used for both production and DW. As a best practice, it is highly recommended to keep the production database on OLTP database and the data warehouse on OLAP.

Also it is observed that the current production database and goldengate is running on an unstable infrastructure, and an immediate attention of experts are recommended to assess the servers, OS, storage layer and network, and do the essential fixes to avoid a catastrophic failure on production environment.

All recommendations are made based on the analysis done on the existing production database environment, and the recommendations are made as per a combination of the best practice of Goldengate deployment and business requirements.